DSA0602 Data Handling and Visualization

Slot C

CO2 – Assessment Tool 3 (AT3)

Case Study Allocation

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- Abstract

This report analyzes retail sales performance using R visualizations to evaluate regional revenue, category contribution, revenue distribution and advertising effectiveness.

Introduction

Retail organizations generate large amounts of transactional data. Visualization transforms raw data into useful insights for decision making.

Problem Statement

Identify the highest-performing region, category contribution, sales distribution and impact of advertising expenditure on sales.

Objectives

1. Compare regional revenue.
2. Analyze category contribution.
3. Study revenue distribution.
4. Evaluate advertising effectiveness.
5. Support managerial decisions.

Dataset Description

The dataset contains Region, Revenue, Advertising Expense, Electronics, Clothing and Grocery revenue.

Methodology

Data Collection → Data Preparation → Visualization → Interpretation → Business Insights

Visualization Selection

Requirement	Visualization
Regional Revenue Comparison	Bar Plot
Category Contribution	Stacked Bar Chart
Revenue Distribution	Histogram
Revenue Density	Density Plot
Advertising vs Sales	Scatter Plot

Dataset Table

Region	Revenue	Advertising	Electronics	Clothing	Grocery
North	150000	25000	70000	50000	30000
South	120000	20000	50000	40000	30000
East	110000	18000	45000	35000	30000
West	130000	22000	60000	45000	25000
Central	100000	15000	40000	30000	30000

Implementation Using R

Figure 1: Regional Revenue Comparison

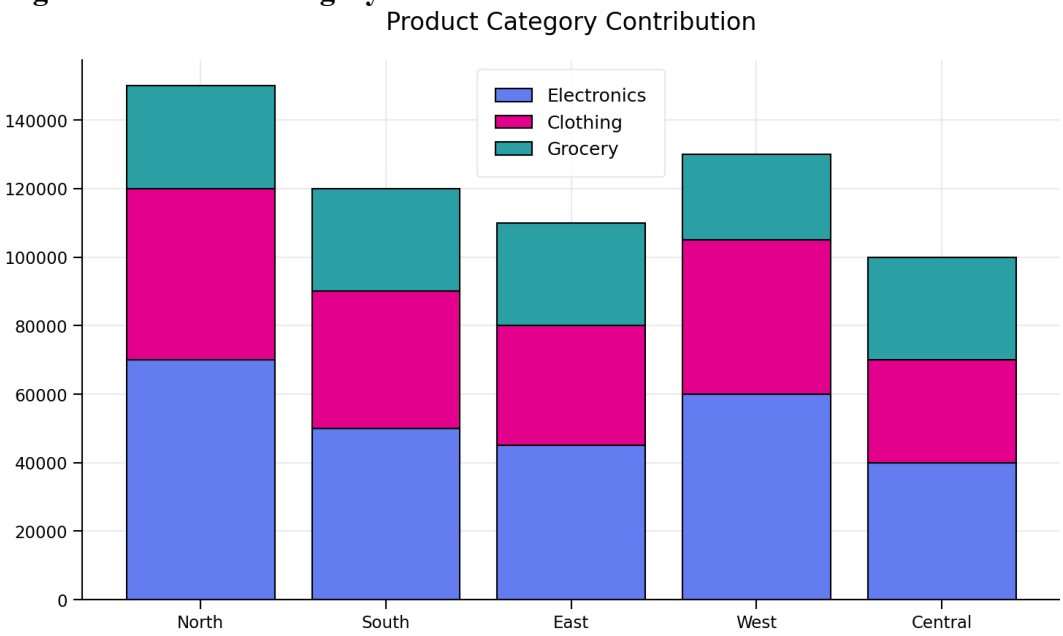


R Code:

```
barplot(sales_data$Revenue, names.arg=sales_data$Region, col="skyblue")
```

Interpretation: North region generated the highest revenue.

Figure 2: Product Category Contribution

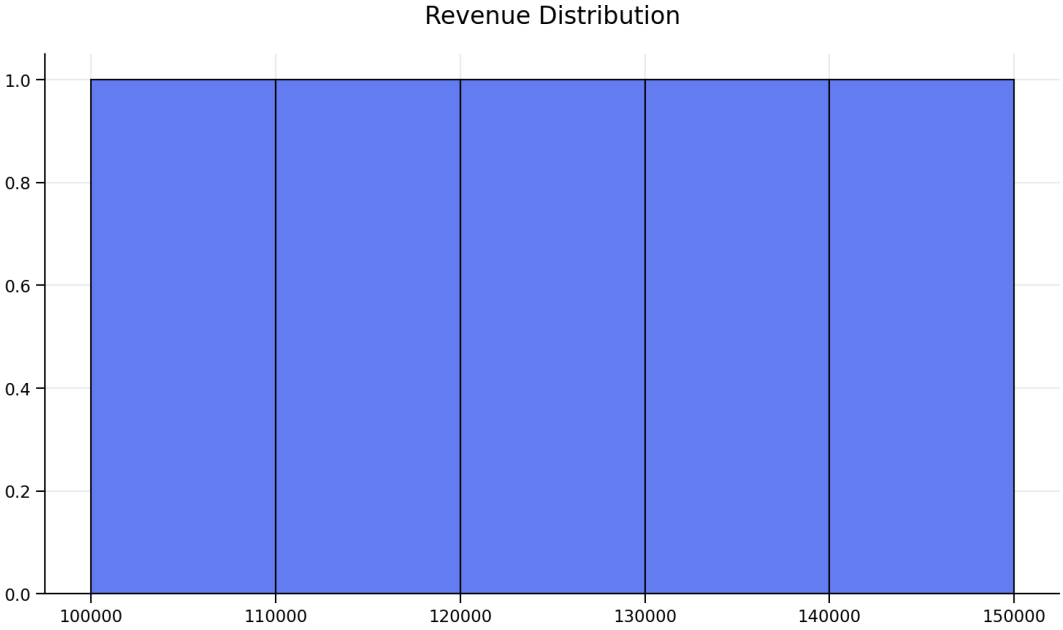


R Code:

barplot(category_data, col=c("red","blue","green"))

Interpretation: Electronics contributes the highest revenue share.

Figure 3: Revenue Distribution

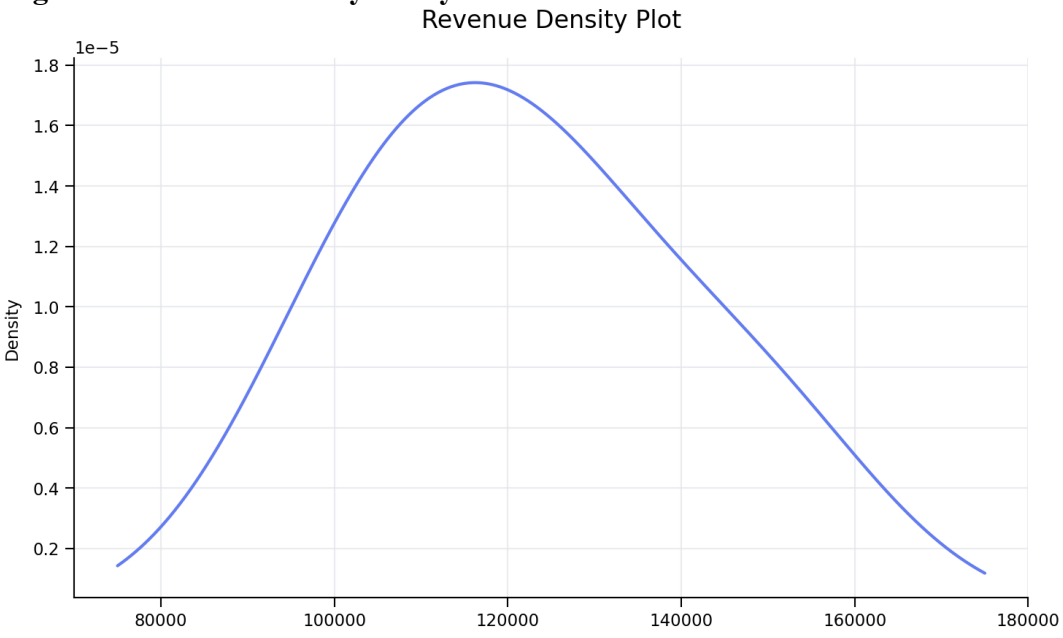


R Code:

hist(sales_data$Revenue, col="lightgreen")

Interpretation: Revenue values are evenly distributed.

Figure 4: Revenue Density Analysis

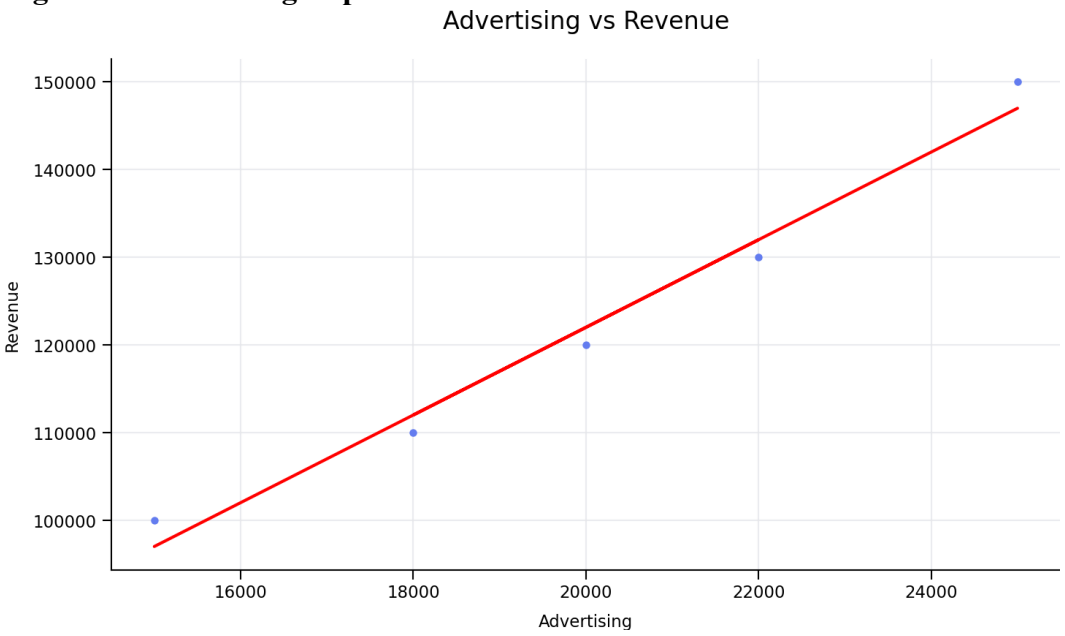


R Code:

plot(density(sales_data$Revenue))

Interpretation: Most revenues cluster around the average.

Figure 5: Advertising Expenditure vs Revenue



R Code:

plot(Advertising_Expense, Revenue)

Interpretation: Positive relationship between advertising and sales.

Results and Interpretation

• Highest Revenue Region: North
• Positive relationship exists between advertising expenditure and sales.
• Electronics contributes the highest percentage of revenue.

Findings

North is the best-performing region and Electronics is the top revenue-generating category.

Conclusion

The dashboard effectively supports business decision-making through data visualization.

Future Scope

Real-time dashboards, predictive analytics and machine learning based forecasting can be incorporated.

References

R Project Documentation
ggplot2 Documentation
Retail Analytics Literature